

analyze_summary_v2.R

Juango the Blue

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```
# Setup -----

# Clear the environment and console
rm(list = ls())
cat("\014")

options(width = 1000)

# Import libraries
library(psych)
library(ggplot2)

##
## Attaching package: 'ggplot2'
## The following objects are masked from 'package:psych':
##
##      %+%, alpha
library(gplots)

##
## Attaching package: 'gplots'
## The following object is masked from 'package:stats':
##
##      lowess

# Import data
results <- read.csv("~/2016-Present (Michigan)/Research/12 Estimation Definitions/04 Analysis/Results/Results.csv")
names(results)[1:7] <- c('index', 'method', 'fn', 'probability', 'cycles', 'performance', 'degree')

# Create vectors of variables
fns = c("sphere", "ackley", "rosen", "stybtang")
methods = c("current", "future", "best")

# Organize Data -----

# Slice data by functions
sphere = results[results$fn == "sphere", c(1:2, 4:7)]
ackley = results[results$fn == "ackley", c(1:2, 4:7)]
rosen = results[results$fn == "rosenbrock", c(1:2, 4:7)]
stybtang = results[results$fn == "styblinski-tang", c(1:2, 4:7)]

# Copy over column names
names(sphere)[1:6] = names(results)[c(1:2, 4:7)]
names(ackley)[1:6] = names(results)[c(1:2, 4:7)]
names(rosen)[1:6] = names(results)[c(1:2, 4:7)]
names(stybtang)[1:6] = names(results)[c(1:2, 4:7)]
```

```

# Slice data by methods
sphere.current = cbind(sphere[sphere$method == "future_est" & sphere$probability == 0,c(1,3:6)],"current")
sphere.future = cbind(sphere[sphere$method == "future_est" & sphere$probability == 1,c(1,3:6)],"future")
sphere.best = cbind(sphere[sphere$method == "best_est" & sphere$probability == 1,c(1,3:6)],"best")
ackley.current = cbind(ackley[ackley$method == "future_est" & ackley$probability == 0,c(1,3:6)],"current")
ackley.future = cbind(ackley[ackley$method == "future_est" & ackley$probability == 1,c(1,3:6)],"future")
ackley.best = cbind(ackley[ackley$method == "best_est" & ackley$probability == 1,c(1,3:6)],"best")
rosen.current = cbind(rosen[rosen$method == "future_est" & rosen$probability == 0,c(1,3:6)],"current")
rosen.future = cbind(rosen[rosen$method == "future_est" & rosen$probability == 1,c(1,3:6)],"future")
rosen.best = cbind(rosen[rosen$method == "best_est" & rosen$probability == 1,c(1,3:6)],"best")
stybtang.current = cbind(stybtang[stybtang$method == "future_est" & stybtang$probability == 0,c(1,3:6)],"current")
stybtang.future = cbind(stybtang[stybtang$method == "future_est" & stybtang$probability == 1,c(1,3:6)],"future")
stybtang.best = cbind(stybtang[stybtang$method == "best_est" & stybtang$probability == 1,c(1,3:6)],"best")

# Rename the new methods column
names(sphere.current)[6] = "method"
names(sphere.future)[6] = "method"
names(sphere.best)[6] = "method"
names(ackley.current)[6] = "method"
names(ackley.future)[6] = "method"
names(ackley.best)[6] = "method"
names(rosen.current)[6] = "method"
names(rosen.future)[6] = "method"
names(rosen.best)[6] = "method"
names(stybtang.current)[6] = "method"
names(stybtang.future)[6] = "method"
names(stybtang.best)[6] = "method"

# Combine the data by method
sphere.methods = rbind(sphere.current,sphere.future,sphere.best)
ackley.methods = rbind(ackley.current,ackley.future,ackley.best)
rosen.methods = rbind(rosen.current,rosen.future,rosen.best)
stybtang.methods = rbind(stybtang.current,stybtang.future,stybtang.best)

# ANOVA -----

# Sphere function
sphere.aov = aov(performance ~ method, data=sphere.methods)
summary(sphere.aov)

##           Df      Sum Sq   Mean Sq F value Pr(>F)
## method      2 1.296e-52  6.483e-53     562 <2e-16 ***
## Residuals 297 3.426e-53  1.200e-55
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

TukeyHSD(sphere.aov)

##      Tukey multiple comparisons of means
##      95% family-wise confidence level
##
## Fit: aov(formula = performance ~ method, data = sphere.methods)
##
## $method

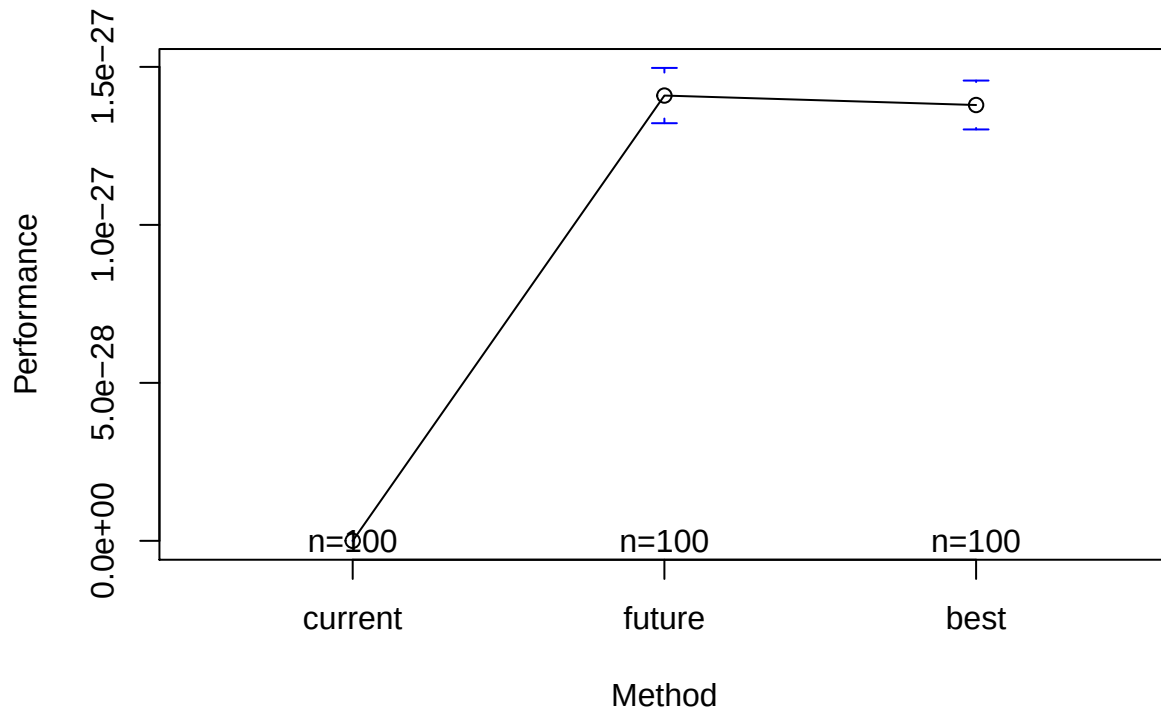
```

```
##               diff               lwr               upr               p adj
## future-current 1.409264e-27 1.296121e-27 1.522406e-27 0.0000000
## best-current   1.379383e-27 1.266241e-27 1.492526e-27 0.0000000
## best-future    -2.988008e-29 -1.430227e-28 8.326253e-29 0.8081361
```

```
plotmeans(performance ~ method,xlab="Method",ylab="Performance",data = sphere.methods)
```

```
## Warning in arrows(x, li, x, pmax(y - gap, li), col = barcol, lwd = lwd, : zero-length arrow is of in
```

```
## Warning in arrows(x, ui, x, pmin(y + gap, ui), col = barcol, lwd = lwd, : zero-length arrow is of in
```



```
# Ackley function
ackley.aov = aov(performance ~ method, data=ackley.methods)
summary(ackley.aov)
```

```
##               Df Sum Sq Mean Sq F value    Pr(>F)
## method          2  16906    8453   7.803 0.000498 ***
## Residuals      297 321723    1083
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
TukeyHSD(ackley.aov)
```

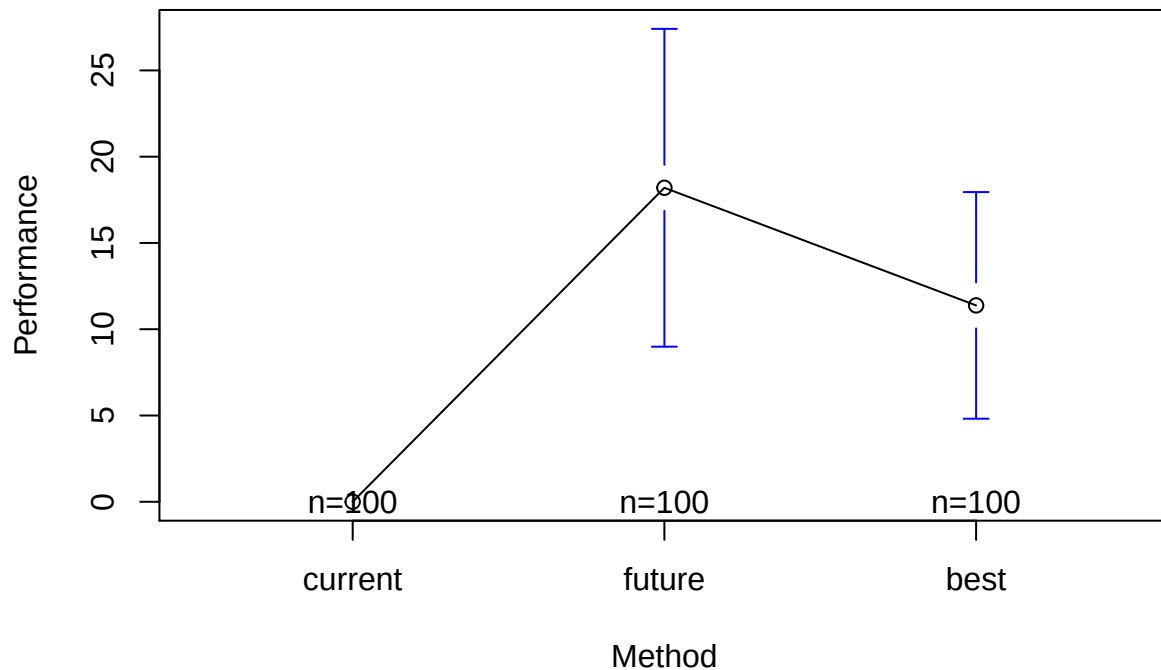
```
## Tukey multiple comparisons of means
## 95% family-wise confidence level
##
## Fit: aov(formula = performance ~ method, data = ackley.methods)
##
## $method
```

```
##           diff      lwr      upr      p adj
## future-current 18.197957  7.2340379 29.161876 0.0003364
## best-current  11.381217  0.4172984 22.345136 0.0398277
## best-future   -6.816739 -17.7806583  4.147179 0.3095401
```

```
plotmeans(performance ~ method,xlab="Method",ylab="Performance",data = ackley.methods)
```

```
## Warning in arrows(x, li, x, pmax(y - gap, li), col = barcol, lwd = lwd, : zero-length arrow is of in
```

```
## Warning in arrows(x, li, x, pmax(y - gap, li), col = barcol, lwd = lwd, : zero-length arrow is of in
```



```
# Rosenbrock function
```

```
rosen.aov = aov(performance ~ method, data=rosen.methods)
summary(rosen.aov)
```

```
##           Df      Sum Sq  Mean Sq F value Pr(>F)
## method      2 3.840e-21 1.922e-21   4.471 0.0122 *
## Residuals 297 1.277e-19 4.299e-22
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

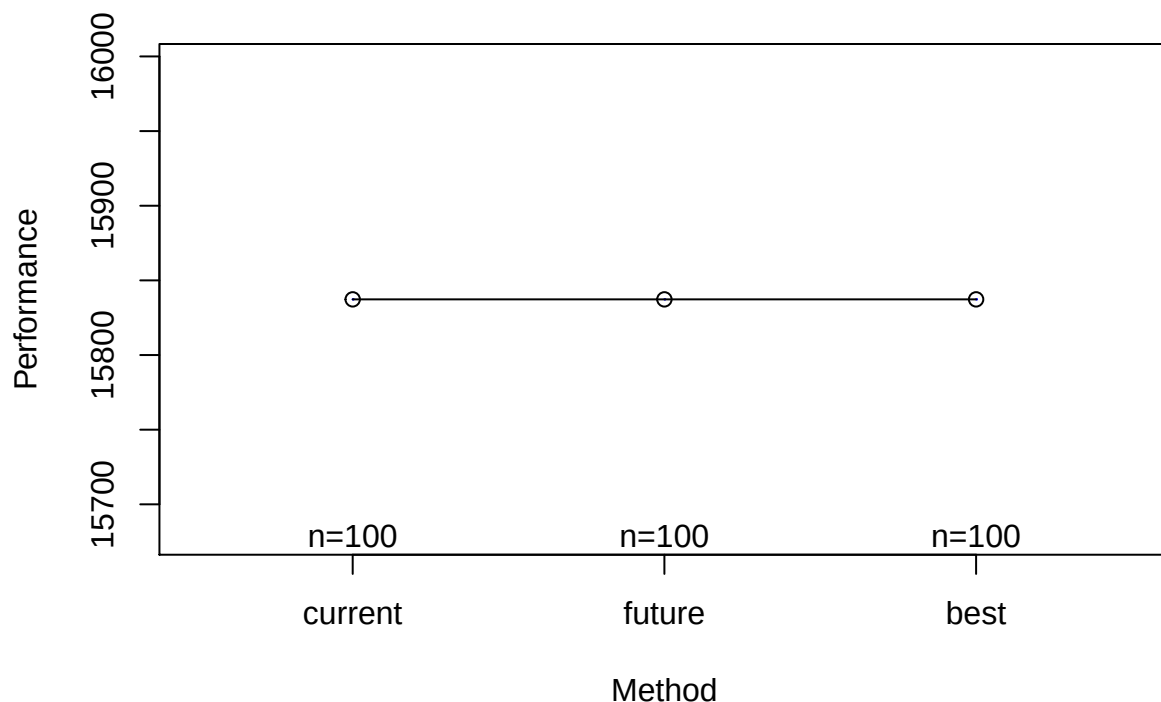
```
TukeyHSD(rosen.aov)
```

```
## Tukey multiple comparisons of means
## 95% family-wise confidence level
##
## Fit: aov(formula = performance ~ method, data = rosen.methods)
##
```

```
## $method
##               diff               lwr               upr               p adj
## future-current -7.275958e-12 -1.418328e-11 -3.686390e-13 0.0362657
## best-current   -7.275958e-12 -1.418328e-11 -3.686390e-13 0.0362657
## best-future     0.000000e+00 -6.907319e-12  6.907319e-12 1.0000000
```

```
plotmeans(performance ~ method,xlab="Method",ylab="Performance",data = rosen.methods)
```

```
## Warning in arrows(x, li, x, pmax(y - gap, li), col = barcol, lwd = lwd, : zero-length arrow is of in
## Warning in arrows(x, li, x, pmax(y - gap, li), col = barcol, lwd = lwd, : zero-length arrow is of in
## Warning in arrows(x, li, x, pmax(y - gap, li), col = barcol, lwd = lwd, : zero-length arrow is of in
## Warning in arrows(x, ui, x, pmin(y + gap, ui), col = barcol, lwd = lwd, : zero-length arrow is of in
## Warning in arrows(x, ui, x, pmin(y + gap, ui), col = barcol, lwd = lwd, : zero-length arrow is of in
## Warning in arrows(x, ui, x, pmin(y + gap, ui), col = barcol, lwd = lwd, : zero-length arrow is of in
```



```
# Styblinski-Tang function
```

```
stybtang.aov = aov(performance ~ method, data=stybtang.methods)
summary(stybtang.aov)
```

```
##           Df    Sum Sq Mean Sq F value Pr(>F)
## method      2 2.460e-20 1.23e-20   1174 <2e-16 ***
## Residuals 297 3.113e-21 1.00e-23
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
TukeyHSD(stybtang.aov)
```

```
## Tukey multiple comparisons of means
## 95% family-wise confidence level
##
## Fit: aov(formula = performance ~ method, data = stybtang.methods)
##
## $method
##              diff              lwr              upr              p adj
## future-current -1.902237e-11 -2.010092e-11 -1.794381e-11 0.0000000
## best-current   -1.939441e-11 -2.047296e-11 -1.831585e-11 0.0000000
## best-future     -3.720401e-13 -1.450595e-12  7.065144e-13 0.6955632
##
## plotmeans(performance ~ method,xlab="Method",ylab="Performance",data = stybtang.methods)
## Warning in arrows(x, li, x, pmax(y - gap, li), col = barcol, lwd = lwd, : zero-length arrow is of in
## Warning in arrows(x, li, x, pmax(y - gap, li), col = barcol, lwd = lwd, : zero-length arrow is of in
## Warning in arrows(x, li, x, pmax(y - gap, li), col = barcol, lwd = lwd, : zero-length arrow is of in
## Warning in arrows(x, ui, x, pmin(y + gap, ui), col = barcol, lwd = lwd, : zero-length arrow is of in
## Warning in arrows(x, ui, x, pmin(y + gap, ui), col = barcol, lwd = lwd, : zero-length arrow is of in
## Warning in arrows(x, ui, x, pmin(y + gap, ui), col = barcol, lwd = lwd, : zero-length arrow is of in
```

